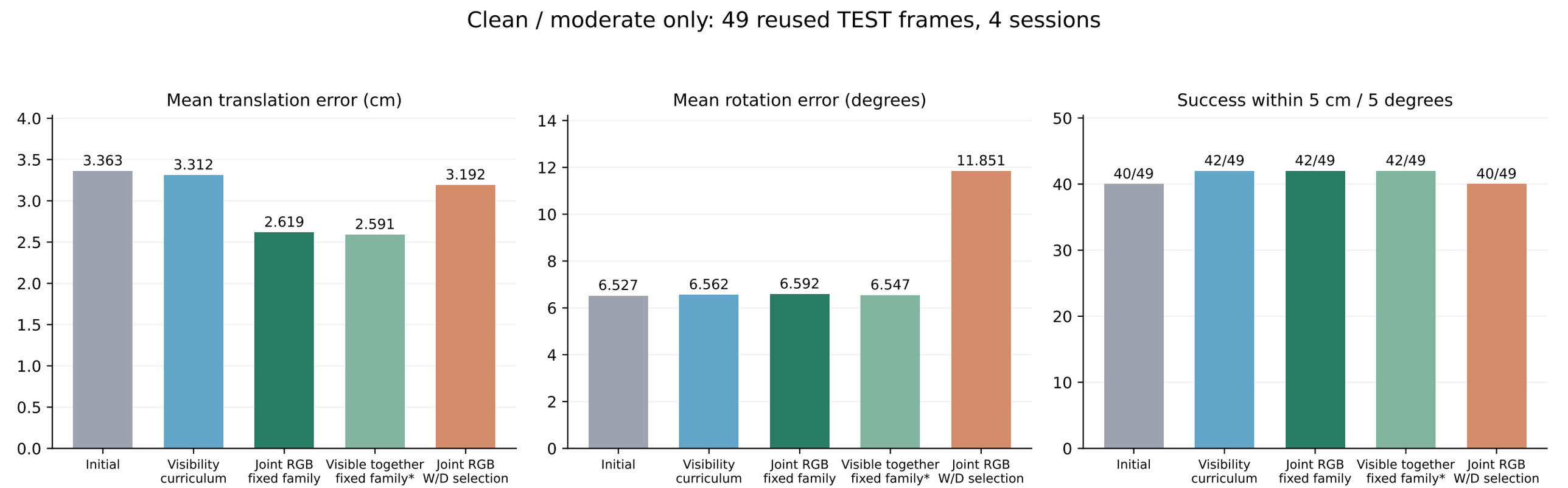
Clean / moderate only: 49 reused TEST frames, 4 sessions
Mean translation error (cm)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
3.363
3.312
2.619
2.591
3.192
Initial
Visibility curriculum
Joint RGB fixed family
Visible together fixed family*
Joint RGB W/D selection
Mean rotation error (degrees)
14
12
10
8
6
4
2
0
6.527
6.562
6.592
6.547
11.851
Initial
Visibility curriculum
Joint RGB fixed family
Visible together fixed family*
Joint RGB W/D selection
Success within 5 cm / 5 degrees
50
40
30
20
10
0
40/49
42/49
42/49
42/49
40/49
Initial
Visibility curriculum
Joint RGB fixed family
Visible together fixed family*
Joint RGB W/D selection
* Same sealed original-family candidates; posthoc comparison. Historical annotation-derived reference; no new physical holdout.